

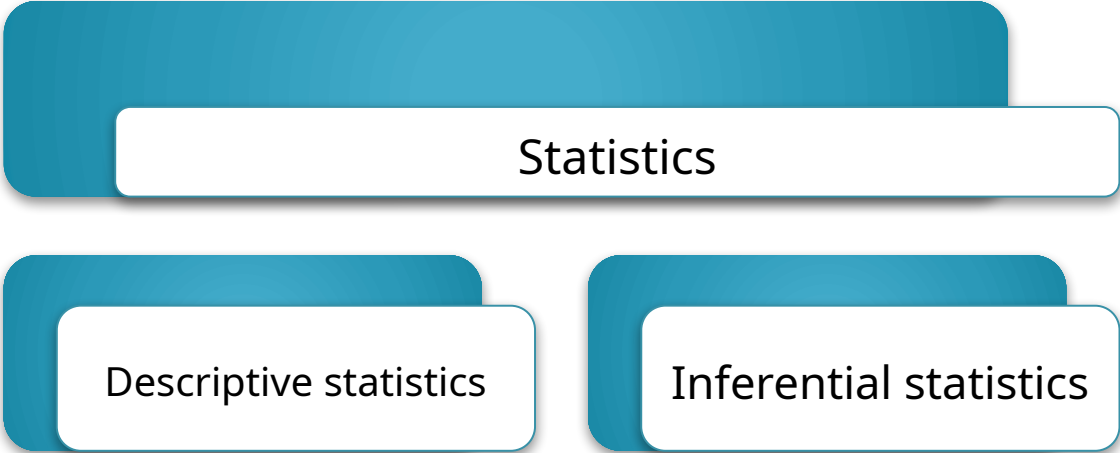
# Parametric vs. non-parametric statistical test

Prakash Chauhan  
CSP15013

# What is statistics?

- A branch of mathematics that provides techniques to analyze whether or not your data is significant(meaningful)
- It may be considered as a branch of mathematics that transform data into useful information for decision makers.

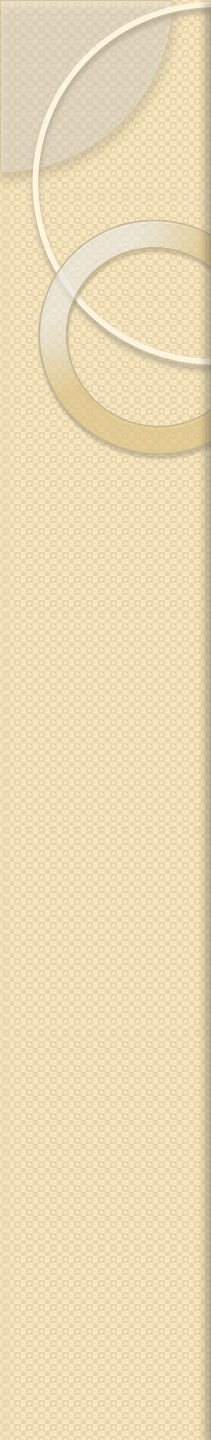
# Types of statistics



Statistics

Descriptive statistics

Inferential statistics



# Descriptive Statistics

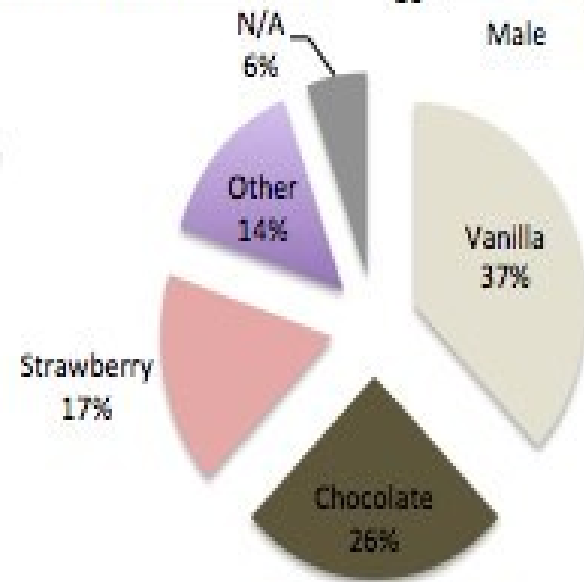
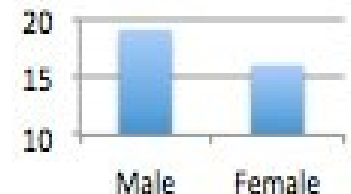
- Collecting, summarizing and describing of data.
- Descriptive statistics do not, however, allow us to make conclusions beyond the data we have analyzed or reach conclusions regarding any hypotheses we might have made.
- They are simply a way to describe our data.

# An example of descriptive statistics

|    | A            | B   | C      | D                         |
|----|--------------|-----|--------|---------------------------|
| 1  | Respondent # | Age | Gender | Favorite Ice Cream Flavor |
| 2  | 1            | 36  | m      | Vanilla                   |
| 3  | 2            | 22  | f      | Chocolate                 |
| 4  | 3            | 61  | m      | Strawberry                |
| 5  | 4            | 88  | m      | Other                     |
| 6  | 5            | 31  | m      | N/A                       |
| 7  | 6            | 53  | m      | N/A                       |
| 8  | 7            | 30  | f      | Chocolate                 |
| 9  | 8            | 64  | f      | Chocolate                 |
| 10 | 9            | 18  | m      | Vanilla                   |
| 11 | 10           | 16  | f      | Vanilla                   |
| 12 | 11           | 83  | m      | Strawberry                |
| 13 | 12           | 16  | f      | Strawberry                |
| 14 | 13           | 94  | m      | Strawberry                |
| 15 | 14           | 55  | m      | Vanilla                   |
| 16 | 15           | 42  | f      | Chocolate                 |
| 17 | 16           | 18  | f      | Vanilla                   |
| 18 | 17           | 21  | f      | Vanilla                   |

Raw Data

| Age           |      |
|---------------|------|
| Mean          | 42.6 |
| Standard Dev. | 21.9 |

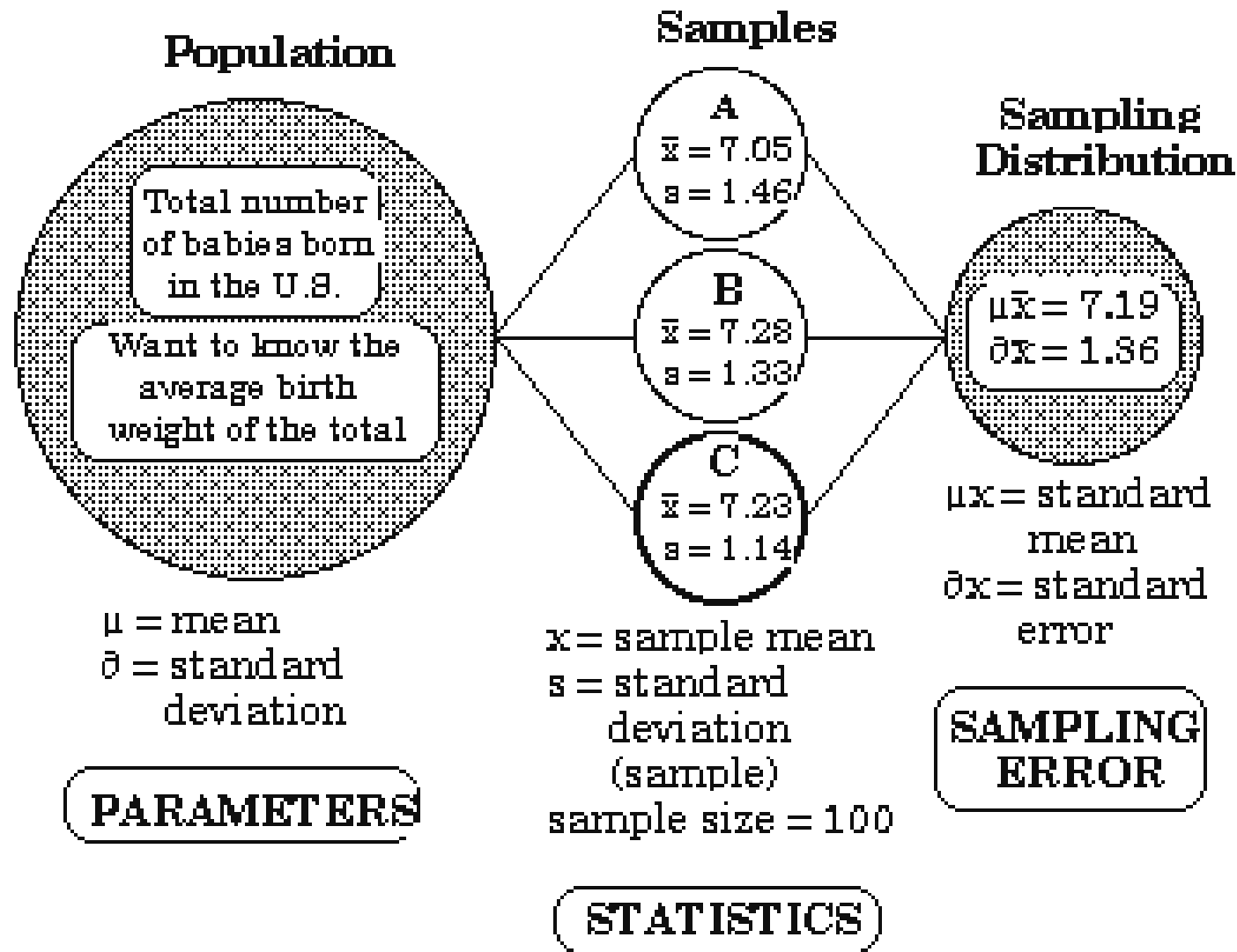


Descriptive Statistics

# Inferential statistics

- It uses
  - ❖ to make inferences
  - ❖ perform hypothesis testing
  - ❖ determining relationships
  - ❖ to make prediction
- Inferential statistics infer from the sample to the population
- They determine probability of characteristics of population based on the characteristics of your sample

# An example of inferential statistics



# Inferential statistics

## Methods of Inferential Statistics

### Estimation Statistics

(used to make estimates about population values based on sample data)

#### Confidence Interval

(provide us with a range of values that we believe, with a given level of confidence, contains a true

Eg: estimating the population mean

#### Parameter Estimation

(allow us to make inferences about how well a particular model might describe the relationship between variables

Eg: linear regression model etc.

### Testing of Statistical Hypotheses

(used to make statistical inferences about whether or not the data we gathered support a particular hypothesis)

#### Parametric Test

(information about the population is completely known by means of parameters)

Eg: t-test, z-test, ANOVA etc.

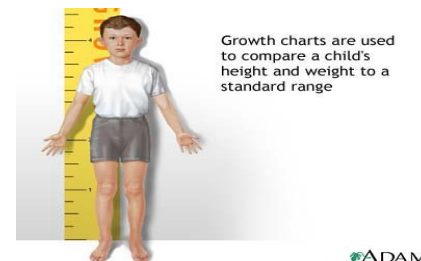
#### Non-Parametric Test

(no knowledge about the population or parameters)

Eg: Mann-Whitney, rank sum test, Kruskal-wallis test etc.

# Types of data (or variable) in statistical analysis

- Cardinal: A cardinal variable tells "how many." Cardinal numbers are also known as "counting numbers," because they show quantity. Eg: 5 apple in a bucket.
- Ordinal: Ordinal variable tell the order of things in a set—first, second, third, etc. Ordinal numbers do not show quantity. They only show rank or position.
- Nominal: A nominal variable names something—a telephone number, a jersey number of a player on a team etc. Nominal numbers do not show quantity or rank. They are used only to identify something.
- Interval: A interval variable is a measurement where the difference between two values is meaningful. Eg: calendar dates
- Ratio : A ratio variable, has all the properties of an interval variable, and also has a clear definition of 0.0. When the variable equals 0.0, there is none of that variable. Eg: height of a kid in meter.



# When and how to choose a parametric or non-parametric test for hypothesis testing?

## ➤ The type of data that you are working with:

- You will typically use a non-parametric test when you are working with data that is nominal or ordinal.
- For interval or ratio data, you may use a parametric test depending on the shape of the distribution

## ➤ The shape of the distribution

- If the measurement scale of your data is interval or ratio, the distribution should be approximately normal to use parametric approaches.
- If you have interval or ratio data with a large sample size ( $> 30$ ) you can usually use parametric approaches due to the Central Limit Theorem.
- Non-parametric tests are more conservative, but less powerful. Often used for small sample sizes ( $< 30$ ), non-normally distributed data.

## ➤ Variances

- Parametric tests often assume homogeneity of variances.
- Non-parametric tests do NOT make assumptions about sample variances

## ➤ Independent and random samples

- Parametric tests assume that samples are independent.
- Non-parametric tests can be used for any type of relationships between samples.

# Parametric vs. Non-parametric test

## Parametric

Information about the population is completely known (Eg: how population are distributed, is it equal-interval etc.)

Specific assumptions are made regarding the population (Eg: distribution is normal)

Null hypotheses is made on parameters of the population distribution (Eg: mean, median, mode, standard deviation etc)

Parametric test is only applicable for interval or ratio data

# Parametric vs. Non-parametric test

## Parametric

Parametric test assumed the homogeneity of variance

Mean is the usual central measure

Parametric test assumed that there is no relationship between data sets (independent)

Can draw more conclusion and as result more powerful if it exist.

# Pros and Cons of parametric and non-parametric test

## ➤ Pros of parametric test:

- The advantage of using parametric tests is their assumptions are more stringent (rigid), thereby making the study results more powerful and requiring less qualification on the part of the researcher in drawing conclusions about the data.

## ➤ Cons of parametric test:

- The main disadvantage of parametric test is that it considered many assumption related to population such as distribution should be normal, data sets are independent, homogeneity in variances, data are continuous etc.

## ➤ Pros of non-parametric test:

- Non-Parametric tests has very less stringent demand on data i.e. it can be used to evaluate data that are not continuous, distribution does not follow normal distribution etc.

## ➤ Cons of non-parametric test:

- Reduces power to detect statistical difference. Eg: For data from a normally distributed population, if the Wilcoxon signed-rank test require 1000 observations to demonstrate statistical significance, a t-test will only require 955.
- Nonparametric tests often require you to modify the hypotheses. For example, most nonparametric tests about the population center are tests about the median instead of the mean. The test does not answer the same question as the corresponding parametric procedure.

# Parametric alternative of non-parametric tests

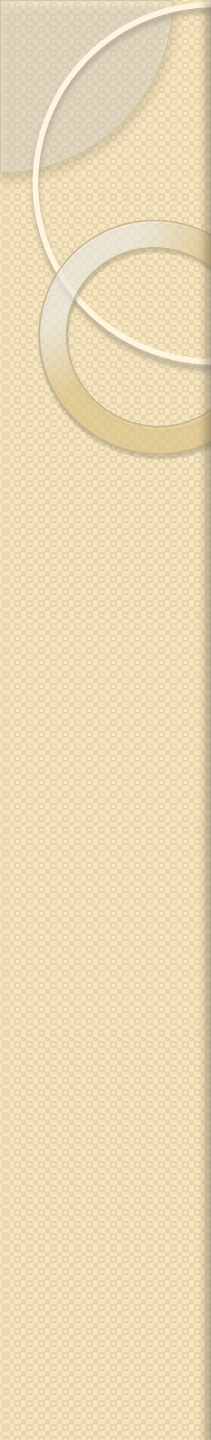
- When a choice exists between using a parametric or a nonparametric procedure, and you are relatively certain that the assumptions for the parametric procedure are satisfied, then use the parametric procedure.
- The following is a list of the nonparametric tests, and their parametric alternatives.

| Nonparametric test     | Alternative parametric test      |
|------------------------|----------------------------------|
| 1-sample sign test     | 1-sample Z-test, 1-sample t-test |
| 1-sample Wilcoxon test | 1-sample Z-test, 1-sample t-test |
| Mann-Whitney test      | 2-sample t-test                  |
| Kruskal-Wallis test    | One-way ANOVA                    |
| Mood's Median test     | One-way ANOVA                    |
| Friedman test          | Two-way ANOVA                    |

# References

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And Few more...



***Thank You !!!***